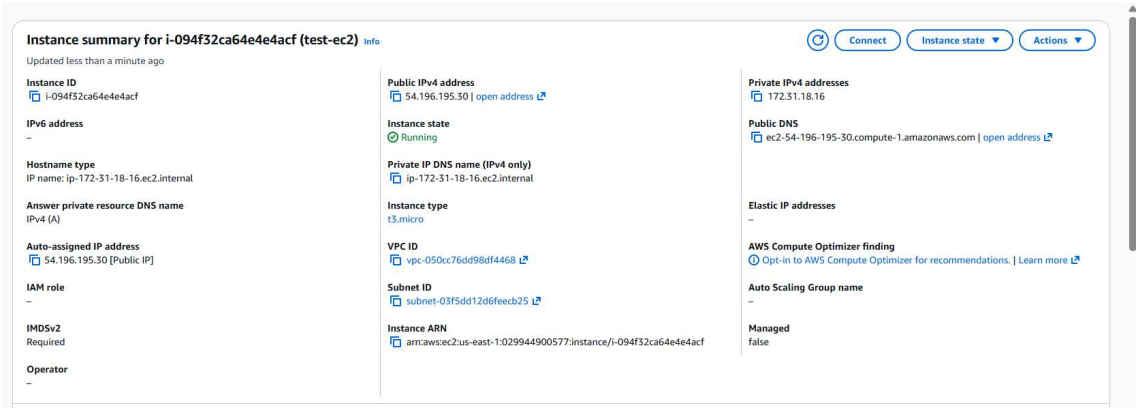
# AWS COST OPTIMIZATION PROJECT

-Project overview
Deleting the stale snapshots using Lambda function(python + boto3)

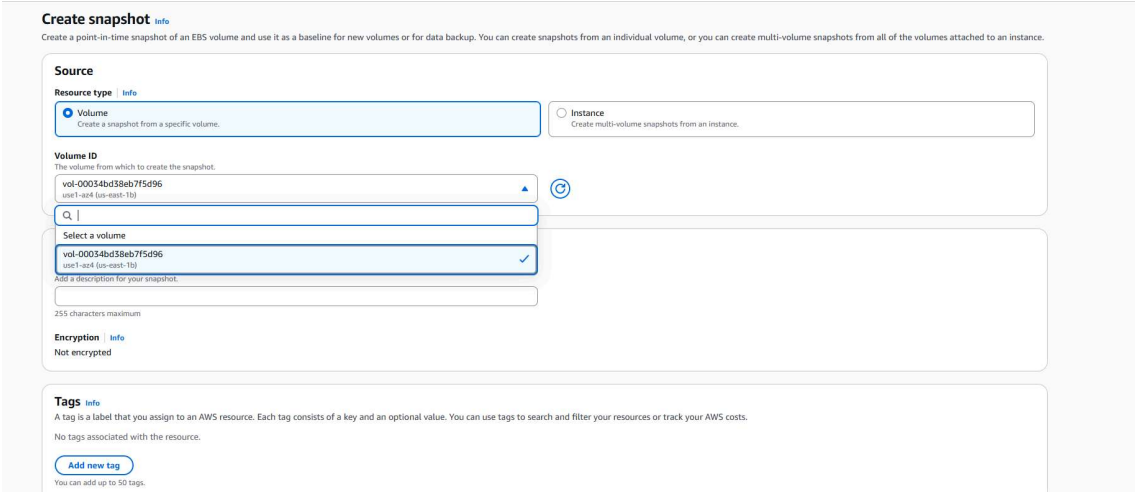
-problem statement
There are some EBS volume Snapshots which are created with ec2 instance but after few days developer deleted ec-2 instance but he forgot to delete the snapshot of that particular volume.
Now using Lambda function we need to delete those stale snapshots which can reduce Aws cost.

-services used in aws
- Ec2 instance
- Lambda function
- IAM Role and policy

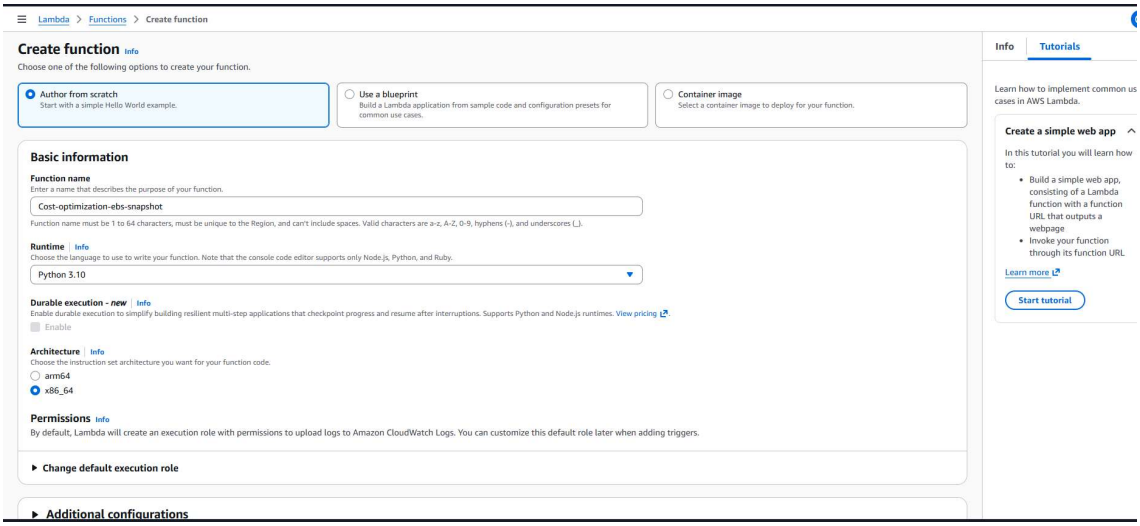
1.Launch a EC-2 instance



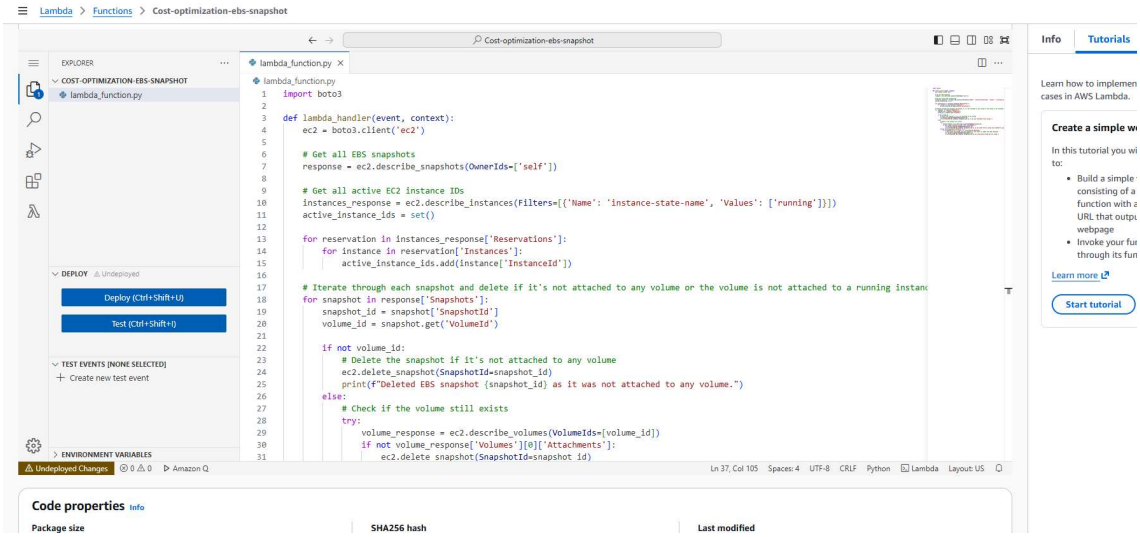
2. Now create a snapshot (just copy of the volume)



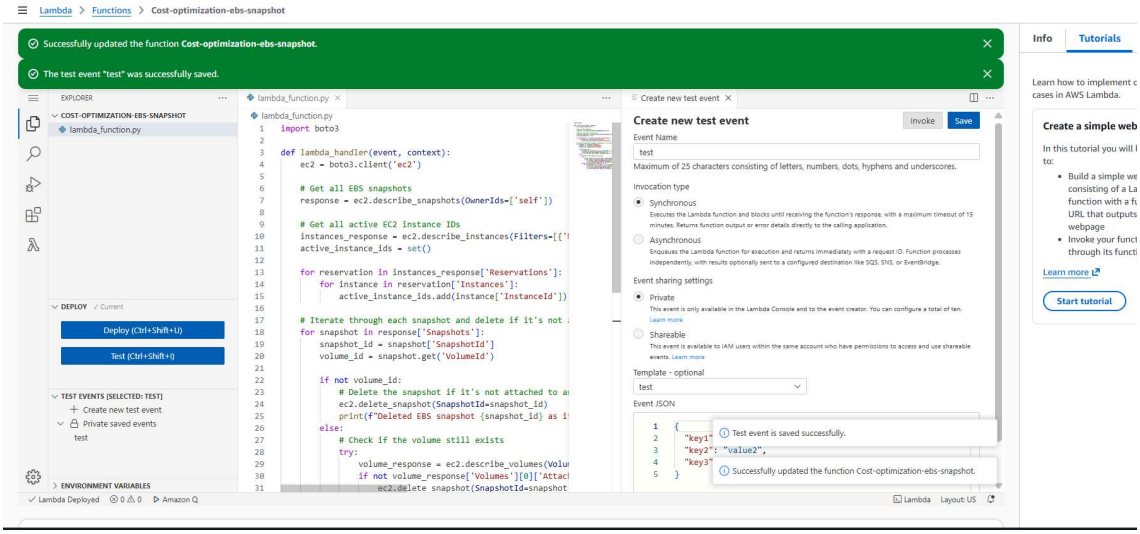

3.create Lambda function



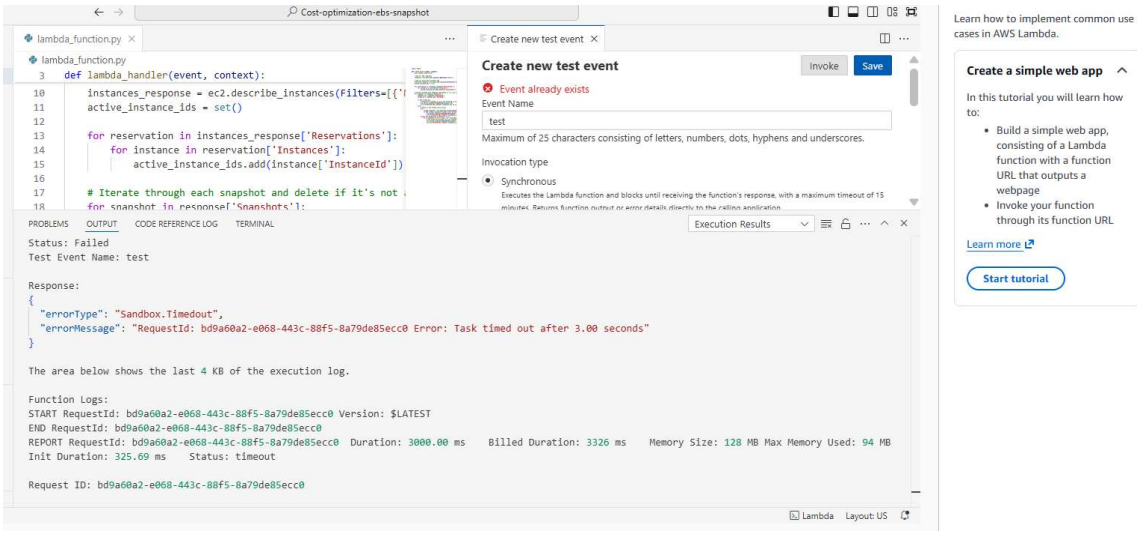
4.insert/write the code and deploy




5.Click on test and create a new test event

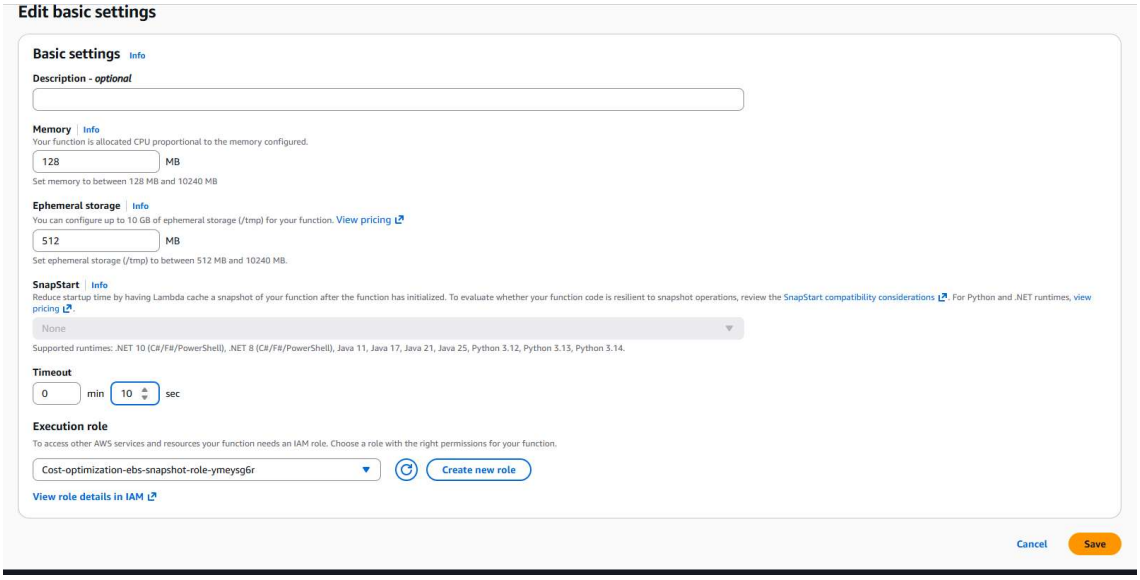


As of now we got some error



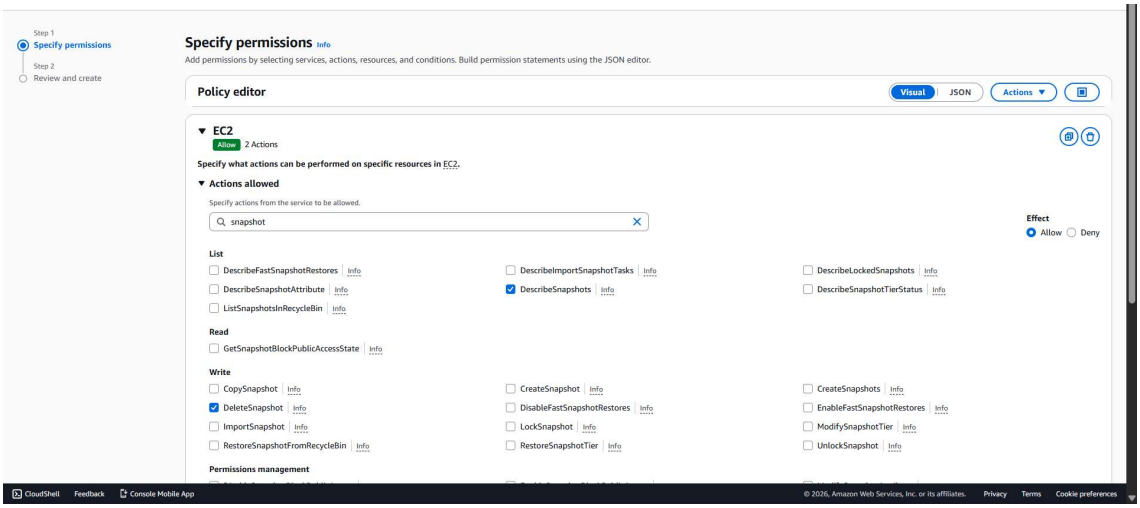

6.now increase the default execution time to 10 seconds

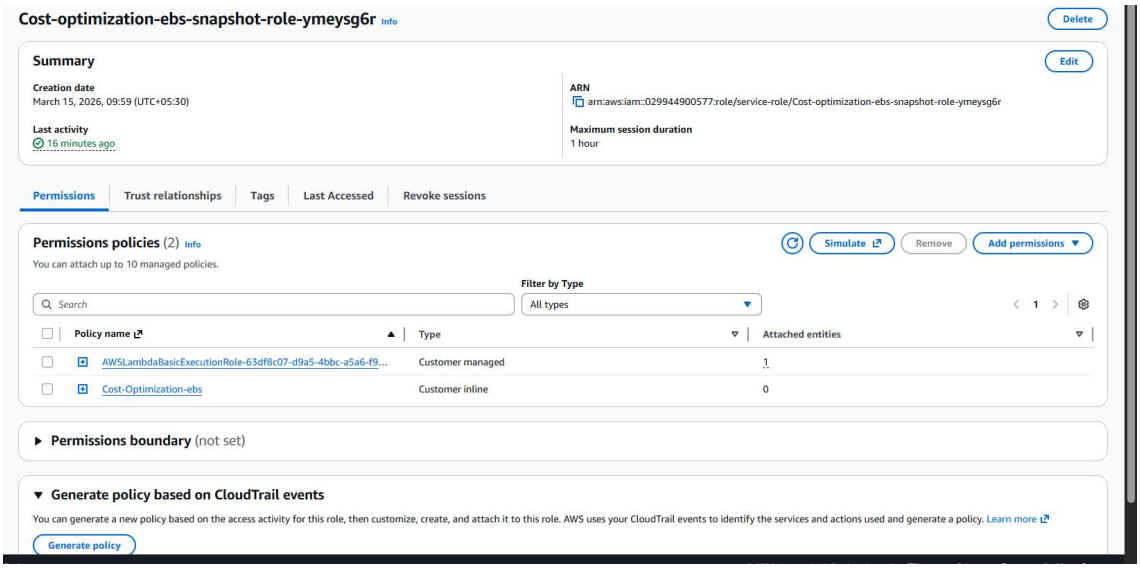
Configuration -> Edit -> change to 10 sec



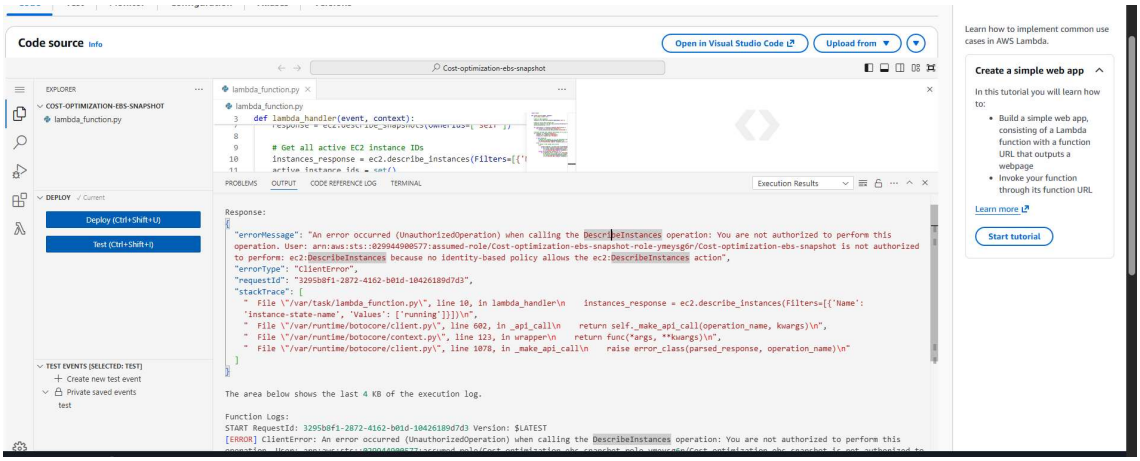
7. Now add the permission
 * describe snapshots
 *delete snapshots
Configuration->permission -> Execution role

8.Now create policy



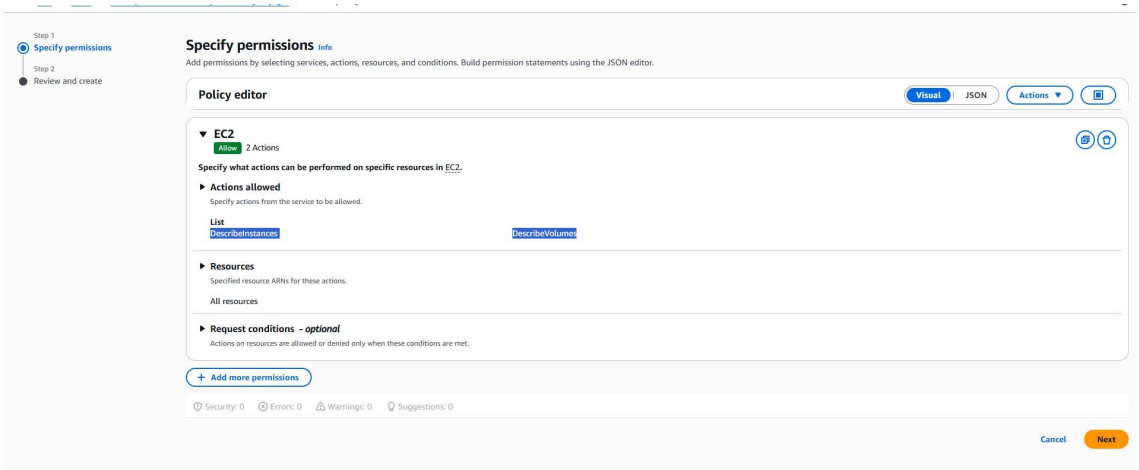



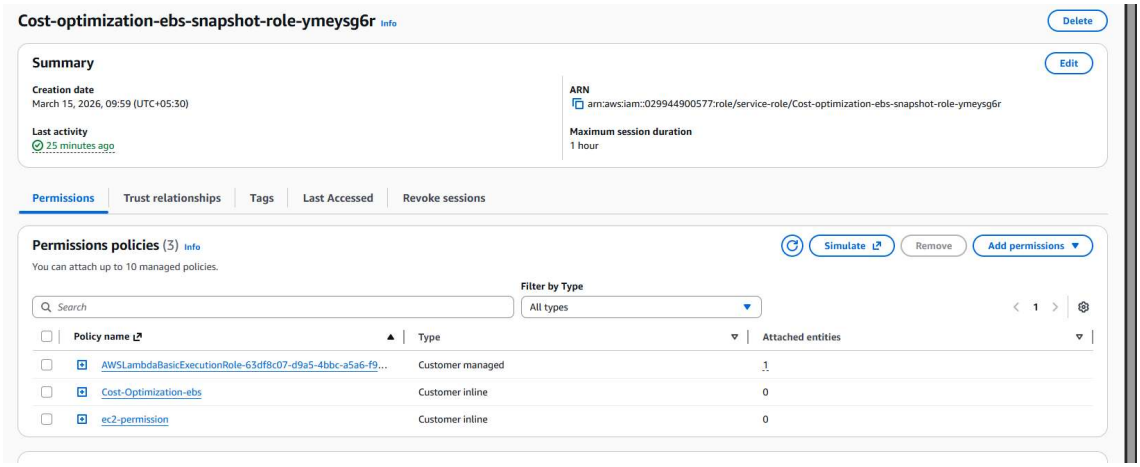
9.now again test the code



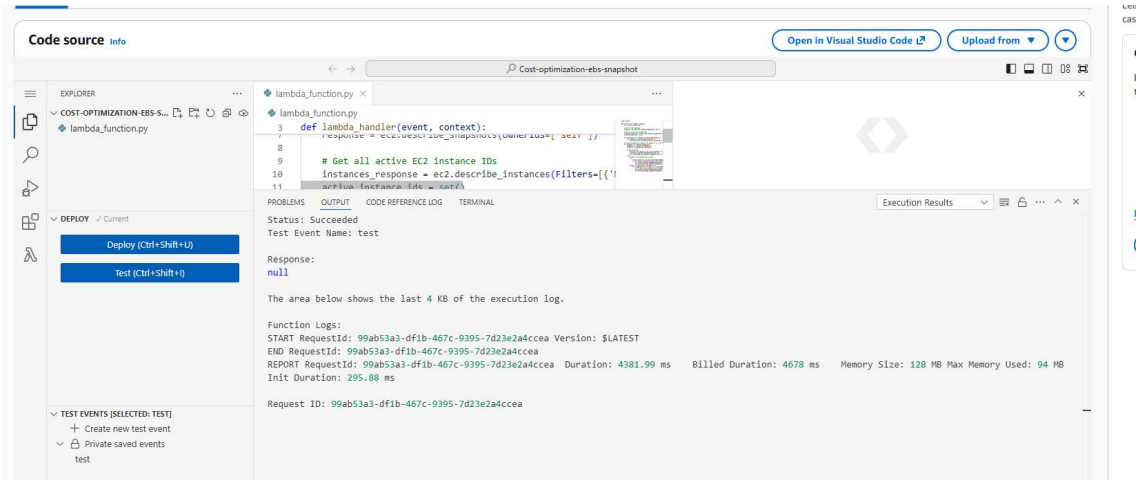

We got few issues

10. Now again create a policy
- Describevolume
- Describe instances



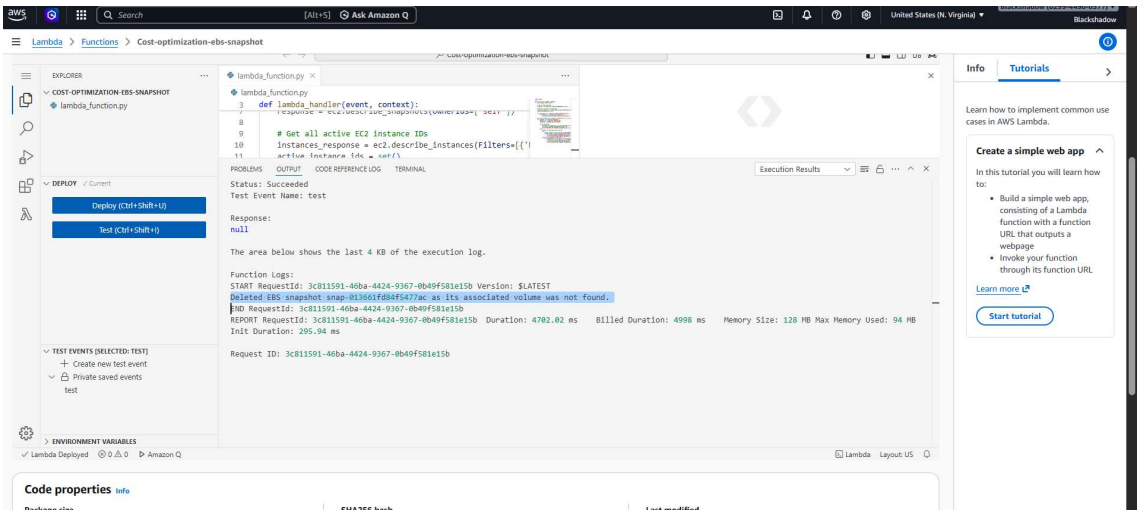
11. Now again test the code




12. Now try to delete the ec-2 instance
NOTE - if u delete the ec2 instance , it will delete the volume also

13. After deleting the ec2 , snapshot is still there

14. Test the code again..




Now we successfully deleted the snapshot which is not associated with any instance.
We can also use the time-stamp and if it is 30 days old …

If a snapshot belongs to a volume that is not attached to a ec-2 instance , then delete the snapshot.
We can change it up to our project requirement.